

1. Administrative & Study Identification

Full Study Title: Phase II trial of neoadjuvant and adjuvant capmatinib in participants with stages IB-IIIA, N2 and selected IIIB (T3N2 or T4N2) NSCLC with MET exon 14 skipping mutation or high MET amplification – Geometry-N **Study Title (Registry):** Phase II of Neoadjuvant and Adjuvant Capmatinib in NSCLC **Short Title/ Acronym:** Geometry-N **Posting ID:** 275017844313864821 **Protocol Number:** CINC280AUS12 **NCT Number:** NCT04926831 **Sponsor/Company Name:** Novartis **Investigational Product:** Capmatinib (INC280) **Trade Name:** Tabrecta **ATC Code:** L01EX17 **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/tabrecta-epar-product-information_en.pdf **Phase of Development:** Phase II **Trial Status:** TERMINATED (Study was terminated early) **Medical Monitor:** Novartis Clinical Operations Lead / Study Director

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine if neoadjuvant capmatinib can improve the major pathological response (MPR) in patients with Stage IB-IIIA, N2 and selected IIIB (T3N2 or T4N2) non-small cell lung cancer (NSCLC) with *MET* exon 14 skipping mutations and/or high *MET* amplification beyond those achieved with surgery, chemotherapy, and radiation.

Secondary Objectives: To evaluate the complete pathological response (pCR) rate, objective response rate (ORR), disease-free survival (DFS), and overall safety and tolerability of the extended adjuvant therapy.

2.2 Background & Rationale Metastatic recurrence is common and associated with poor survival in early-stage (stage I-IIIB) NSCLC despite standard of care treatments like surgery and chemoradiation. *MET* exon 14 skipping mutations and high-level *MET* amplification are actionable oncogenic drivers. Capmatinib is a highly selective *MET* inhibitor approved for advanced *MET*ex14 NSCLC. This trial evaluates the efficacy and safety of neoadjuvant and adjuvant capmatinib to prevent recurrence and improve clinical outcomes in this specific genetically defined patient population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Multi-cohort, parallel group assignment within cohorts without a placebo control arm) **Blinding:** Open-label

Randomization: N/A (Parallel assignment to Cohort A or Cohort B based on mutation/amplification status)

3.2 Planned Enrollment & Duration Targeted Enrollment: 4

participants. (Note: Approximately 42 participants were initially aimed to be enrolled to obtain 38 evaluable participants. The study recruitment was prematurely discontinued due to significant recruitment challenges leading to termination of the study. Challenges included the rarity of the mutation and site initiation delays caused by staff shortages associated with COVID-19. As a result, only 4 subjects were enrolled and treated). **Number of Sites:** Multicenter **Estimated Study Start:** 30-Sep-2021 **Estimated Primary Completion:** 31-Mar-2028 (Expected initially prior to termination)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult ≥ 18 years of age at the time of informed consent. 2. Histologically confirmed NSCLC Stage IB-IIIA, N2 and selected IIIB (T3N2 or T4N2) (per AJCC 8th edition), deemed suitable for primary resection by treating surgeon (T4 tumors with mediastinal organ invasion were not eligible for enrollment). 3. Participant must have *MET* exon 14 skipping mutation and/or high-level *MET* amplification (*MET*: GCN ≥ 10) as determined by a CLIA certified laboratory. High level *MET* amplification must be identified by FISH in a CLIA certified laboratory or FoundationOne CDx NGS (other NGS-based methods without adjusting for tumor content % cannot be accepted). 4. Participants must be eligible for surgery and scheduled for surgical resection within approximately 2 weeks after the last dose of neoadjuvant study treatment. 5. Eastern Cooperative Oncology Group (ECOG) performance status (PS) of 0 or 1.

4.2 Exclusion Criteria 1. Participants with unresectable or metastatic disease. All participants should have brain imaging (either MRI brain or CT brain with contrast) prior to enrollment to exclude brain metastasis. 2. Presence or history of a malignant disease that has been diagnosed and/or required therapy within the past 3 years. Exceptions to this exclusion include the following: completely resected basal cell and squamous cell skin cancers, and completely resected carcinoma in situ of any type. 3. Prior treatment with any *MET* inhibitor or HGF-targeting therapy. 4. Participants with other known oncogenic driver alterations. 5. Prior systemic anti-cancer therapy (chemotherapy, immunotherapy, biologic therapy, vaccine) or investigational agents for NSCLC. 6. Participants with known hypersensitivity to capmatinib and any of the excipients of capmatinib (crospovidone, mannitol, microcrystalline cellulose, povidone, sodium lauryl sulfate, magnesium stearate, colloidal silicon dioxide, and various coating premixes).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Capmatinib 400 mg twice daily. **Route of Administration:** Oral **Duration of Treatment:** 8 weeks (2 cycles) of neoadjuvant therapy prior to surgical resection, followed by up to 3 years of adjuvant therapy.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care and post-operative management. **Prohibited:** Prior or concurrent treatment with any other MET inhibitor, HGF-targeting therapy, or other systemic anti-cancer therapies for NSCLC.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -1	Informed Consent, MET molecular testing (CLIA/FoundationOne CDx), baseline imaging, ECOG PS
Neoadjuvant Phase	Week 1 to 8	Capmatinib 400 mg BID for 2 cycles, pre-surgical safety labs, tumor assessment
Surgery	Within ~14 Days post-treatment	Surgical resection of tumor, pathological tissue assessment for MPR/pCR
Adjuvant Phase	Up to 3 Years	Restart capmatinib 400 mg BID, regular safety and DFS monitoring
End of Study	Post-treatment	2-year survival follow-up period

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Major Pathological Response (MPR) rate, defined as the proportion of participants with $\leq 10\%$ residual viable cancer cells. **Measurement Method:** Local assessment of resected tumor tissue post-neoadjuvant therapy. *Note: Due to the low number of participants and limited data resulting from early termination, analysis related to the primary endpoint was not performed.*

7.2 Secondary & Exploratory Endpoints * Overall Response Rate (ORR) * Complete Pathologic Response (pCR) Rate * Disease Free Survival (DFS)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard Treatment-Emergent Adverse Event (TEAE) monitoring starting from the first dose until 30 days after the last dose of study treatment. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent data monitoring and audits performed per Novartis standard Quality Management System.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set and Efficacy Evaluable Set separated by cohorts (Cohort A: *MET*ex14; Cohort B: *MET*amp).

Sample Size Justification: A 2-stage design was planned. Stage 1 required 9 patients per cohort to evaluate MPR; Stage 2 was to enroll 10 more patients if ≥ 1 of 9 participants achieved an MPR. Due to study termination, target enrollment was not met.

Handling of Missing Data: Standard time-to-event survival analysis (Kaplan-Meier estimates) for DFS and descriptive statistics for tumor response endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Risk-based monitoring and auditing of investigator sites according to Novartis SOPs. **Audit Trail:** eCRF tracking, centralized pathological review verification, and data clarification forms via the Novartis Quality Management System (QMS).

11. Study Identifiers & Governance

Posting ID: 275017844313864821 **Primary ID:** CINC280AUS12 **Registry Number:** NCT04926831 **Sponsor/Lead Organization:** Novartis **Study Title:** Geometry-N / Phase II of Neoadjuvant and Adjuvant Capmatinib in NSCLC **Official Regulatory Title:** Phase II trial of neoadjuvant and adjuvant capmatinib in participants with stages IB-IIIA, N2 and selected IIIB (T3N2 or T4N2) NSCLC with *MET* exon 14 skipping mutation or high *MET* amplification (Geometry-N)

12. Clinical Framework (Study Specific)

Disease Area: Non-small Cell Lung Cancer **Primary Condition:** Metastatic Non Small Cell Lung Cancer **UMLS Preferred Name:** Non-small Cell Lung Carcinoma **Condition Preferred Name:** Metastatic non-small cell lung cancer **MeSH Code:** D016938 **SNOMED ID:** 457721000124104 **Diagnosis Threshold:** *MET* exon 14 skipping mutation or high *MET* amplification (Gene Copy Number ≥ 10) **Medical Methodology:** Neoadjuvant targeted therapy followed by surgery and adjuvant therapy **Phase Requirement:** Trial must be in PHASE2 **Optimization Target:** Major Pathological Response (MPR) and extending Disease-Free Survival (DFS)

13. Study Procedures & Methodology

Management Pathway: Targeted oral TKI therapy prior to and following primary surgical resection **Education Protocol (HCP):** Identification of high-risk actionable genomic alterations via CLIA-certified NGS/FISH assays **Education Protocol (Patient):** Informed consent, surgical readiness, and TKI adherence counseling **Quality Audit Mechanism:** Local and central dual-review of pathological response (MPR and pCR)

14. Observation & Follow-Up Schedule

Total Duration: 8 weeks neoadjuvant + up to 3 years adjuvant + 2 years survival follow-up **Onsite Visit Cadence:** Baseline, end of neoadjuvant cycles (Weeks 4 and 8), perioperative period, and scheduled intervals during the 3-year adjuvant phase **Remote Monitoring Frequency:** Standard safety check-ins as per oncology trial protocols **Checklist Review Interval:** End of each treatment cycle for tumor assessments and safety labs

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				